

Antoine EDY

AI Research Scientist

+33 6 75 36 89 17
antoineedy@outlook.fr

antoine-edy | Google Scholar | antoineedy | antoineedy

Education

- **University of Surrey** — *Master of Science in Artificial Intelligence* *2023-2024*
  - International Excellence Scholarship recipient. Graduated with Highest Honor.
  - Master Thesis: *Zero-shot and Few-shot Adaptation of Multimodal Foundation Models*, supervised by Dr Muhammad Awais Rana
- **École Centrale de Lyon** — *General Engineering Program (Grande École)* *2021-2024*
  - Specializations: Machine Learning, Probability Theory, Chaos and Fractals
  - Project Leader: Unsupervised MCU Classification with STMicroelectronics
- **Collège Stanislas Paris VI** — *Preparatory Classes (MPSI-MP), Mathematics & Physics* *2019-2021*

Publications

- **ViDoRe V3: A Comprehensive Evaluation of RAG in Complex Real-World Scenarios**
  - A. Loison*, Q. Macé*, **A. Edy***, V. Xing, T. Balough, G. Moreira, B. Liu, M. Faysse et al.
  - Multimodal RAG benchmark: 26K pages, 3,099 queries across 10 domains in 6 languages, 12,000h of human annotation
  - **ACL 2026 Main Conference**, arXiv:2601.08620
- **Working Notes on Late Interaction Dynamics: Analyzing Targeted Behaviors**
  - **A. Edy***, M. Conti*, Q. Macé
  - Analysis of length bias in multi-vector scoring and MaxSim similarity distribution on NanoBEIR benchmark
  - **ECIR 2026** Late Interaction Workshop, arXiv:2603.26259
- **Exploring Zero-Shot Capabilities of Multi-Modal Foundation Models for Semantic Segmentation**
  - Improved ZegCLIP baseline using multi-scaling techniques and inner encoder layers for zero-shot segmentation
  - **Master's Thesis**, University of Surrey, ResearchGate

Research & Industry Experience

- **Illuin Technology** — *AI Research Scientist* *September 2024 - Present*
  - Co-led the creation of **ViDoRe V3**, establishing the gold standard benchmark for multimodal document retrieval evaluation; contributor to the **MTEB** ecosystem
  - Conducted research on late interaction dynamics (ColBERT/ColPali-like models): identified and characterized length bias in multi-vector scoring, published first-author paper at ECIR 2026
  - Fine-tuned LLMs for tool calling in agentic RAG contexts; fine-tuned speech-to-text models (Whisper, NeMo) for domain-specific industrial applications
  - Designed and implemented the ViDoRe V3 Pipeline evaluation framework and leaderboard for end-to-end retrieval pipeline comparison
- **Centre for Vision, Speech and Signal Processing (University of Surrey)** — *Research Intern* *2024*
  - Studied zero-shot semantic segmentation using multimodal foundation models; improved ZegCLIP baseline with multi-scaling techniques and inner encoder layers
  - Research led to Master's thesis: *Zero-shot and Few-shot Adaptation of Multimodal Foundation Models*, supervised by Dr Muhammad Awais Rana
- **AI-vidence** — *ML Research Intern* *2023*
  - Developed a python library for ML/AI model explainability (SHAP, LIME, Xplique)
  - Presented at SpringTech Paris-Saclay 2023 (HEC)

Talks & Presentations

- **ViDoRe V3 and Visual Retrieval** — **Neurons & Peppers**, Invited talk at Datacraft Paris *2026*
- **Late Interaction Dynamics** — **Poster presentation**, ECIR 2026 Late Interaction Workshop *2026*

Technical Skills

Deep Learning: PyTorch, Hugging Face (Transformers, PEFT, Datasets, Evaluate), NVIDIA NeMo, OpenMMLab
ML & Data: Scikit-learn, Pandas, NumPy, OpenCV, Seaborn, Jupyter
Retrieval & NLP: ColBERT, ColPali, MTEB, Sentence-Transformers, BM25, FAISS
Explainability: SHAP, LIME, LENS, Xplique
Languages: Python (advanced), C/C++, SQL, Bash, LaTeX
Spoken Languages: French (Native), English (Fluent — C1, TOEFL 623/677), Spanish (B1)